
CiA 444 implementation guideline for CANopen

March 23, 2021



Foreword

CAN in Automation (CiA) e. V. is a nonprofit users' and manufacturers' group. The work of preparing CiA technical documents is normally carried out through a CiA technical group (e.g. Interest Groups or Special Interest Groups). Each CiA member, interested in a subject for that such a group has been established, has the right to be represented on that group.

The procedures used to develop this document and those intended for its further maintenance are described in the CiA rules. This document was drafted in accordance with editorial rules similar to those used by ISO and IEC.

CiA draws attention to the possibility that the implementation of this document can involve the use of (a) patent(s). CiA takes no position concerning the evidence, validity, or applicability of any claimed patent rights in respect thereof. CiA lists reception of notice of patents and associated license declarations in the CiA patent database (www.can-cia.org/s/patents). However, implementers are cautioned that this cannot represent the latest information. CiA shall not be held responsible for identifying any or all such patent rights.

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement. CANopen, CANopen FD, CANsec, and CiA are European Union trademarks which are registered by CiA e. V.; CAN XL is a trademark of Robert Bosch GmbH. The use of them is permitted for CiA members. Registered parties may use CANopen and CANopen FD. More detailed terms for the use of these European Union trademarks are available from CiA.

There is no warranty for this document, to the extent permitted by applicable law. Except when otherwise stated in writing the copyright holder and/or other parties provide this document "as is" without warranty of any kind, either expressed or implied, including, but not limited to, the implied warranties of merchantability and fitness for a particular purpose. The entire risk as to the correctness and completeness of the specification is with you. Should this document prove failures, you assume the cost of all necessary servicing, repair or correction.

CiA is committed to use inclusive language in its specifications and technical reports. CiA documents are going to be updated with the inclusive terms. A listing of inclusive terms and terms they substitute is provided in the CiA house style document downloadable from the CiA website (www.can-cia.org/s/housestyle).

NOTE Please visit the CiA website for the most recent version of this document.

This version replaces Version 1.0.0, which has been revised editorially. The main changes are as follows:

- Minor editorial improvements;
- Removing permissions; and
- Adding of Foreword and Introduction sections.

© CiA 2021

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from CiA at the address below.

CAN in Automation e. V.
Kontumazgarten 3
DE – 90429 Nuremberg, Germany
Phone.: +49-911-928819-0
Url: www.can-cia.org
Email: headquarters@can-cia.org

Introduction

Goal of this document is to explain the different ways to physically connect spreader (or other crane add-on devices) to a crane with or without CANopen interfaces. It is intended for system designers.

Both, crane and spreader can be directly wired or equipped with a CANopen interface. In the normal application of a crane it is possible to have different spreaders attached to different cranes. The four possible configurations are:

- CAN connection between a new crane and a new spreader
- CAN or direct-wired connection between an old crane and a new spreader
- CAN or direct-wired connection between a new crane and an old spreader
- Direct-wired connection between an old crane and an old spreader

CAN connection between a crane and a spreader can be performed as shown in Figure 1.

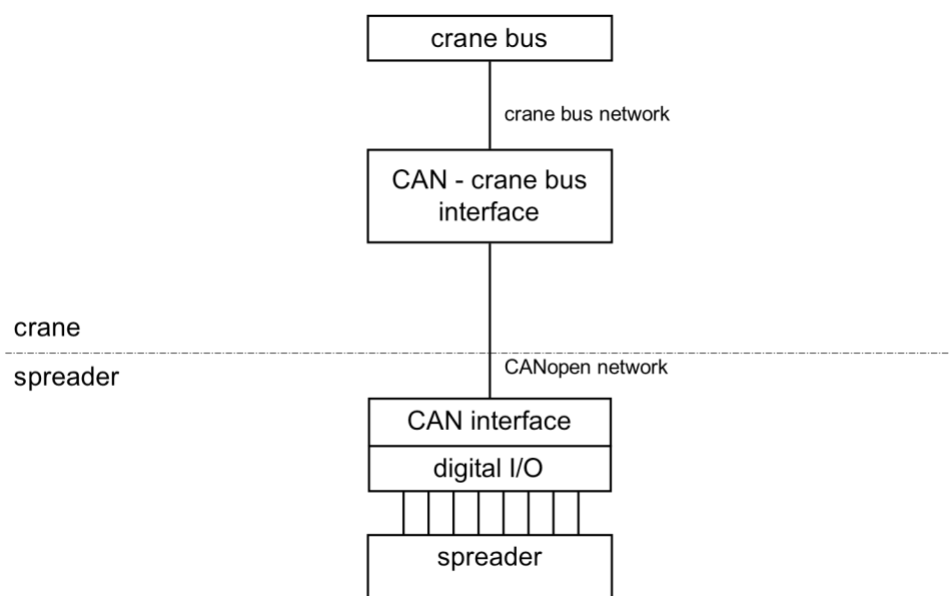


Figure 1 – CAN connection between a crane and a spreader

Contents

1	Scope	5
2	References.....	5
3	Definitions	5
4	CAN or direct-wired connection between an old crane and a new spreader	5
5	CAN or direct-wired connection between a new crane and an old spreader	6
6	Direct-wired connection between an old crane and an old spreader	7
6.1	General	7
6.2	Hardware compatibility	7
6.3	Manufacturer-specific transmission with a boosted CANopen network	7

1 Scope

This document describes the recommended practice and gives application hints for implementing the connection of crane and spreader compliant with the CiA 444-1 and CiA 444-2 specifications.

2 References

The following referenced documents are indispensable for the application of this document. For undated references, the latest edition of the referenced document (including any amendments) applies.

CiA 444-1, CANopen profile for container-handling machine add-on devices – Part 1: General definitions

CiA 444-2, CANopen profile for container-handling machine add-on devices – Part 2: Spreader for crane

ISO 11898-2, Road vehicles – Controller area network (CAN) – Part 2: High-speed medium access unit

3 Definitions

CiA, IEC, and ISO maintain terminological databases for use in specifications respectively standards at the following addresses:

- CiA glossary of terms: <https://www.can-cia.org/groups/cia-glossary-of-terms/>
- IEC Electropedia: <http://www.electropedia.org/>
- ISO Online browsing platform: <https://www.iso.org/obp/>

For the purposes of this document, the following definitions and those given in CiA 444-1, CiA 444-2, and ISO 11898-2 apply.

new crane

crane with a CANopen interface

new spreader

spreader with a CANopen interface

old crane

crane without a CANopen interface

old spreader

spreader without a CANopen interface

4 CAN or direct-wired connection between an old crane and a new spreader

Figure 2 shows the CAN or direct-wired connection between an old crane and a new spreader.

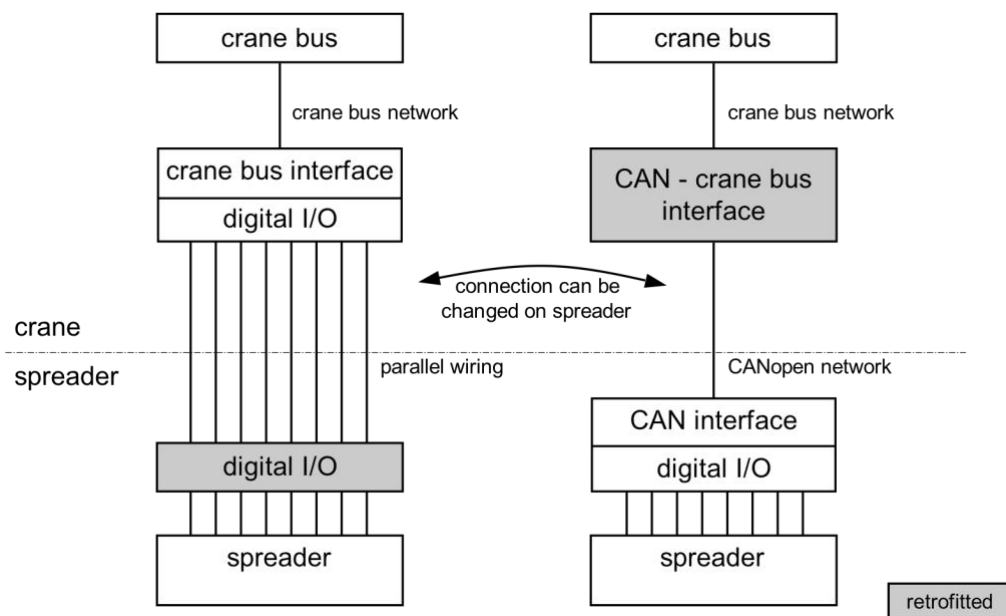


Figure 2 – CAN or direct-wired connection between an old crane and a new spreader

5 CAN or direct-wired connection between a new crane and an old spreader

Figure 3 shows the CAN or direct-wired connection between a new crane and an old spreader.

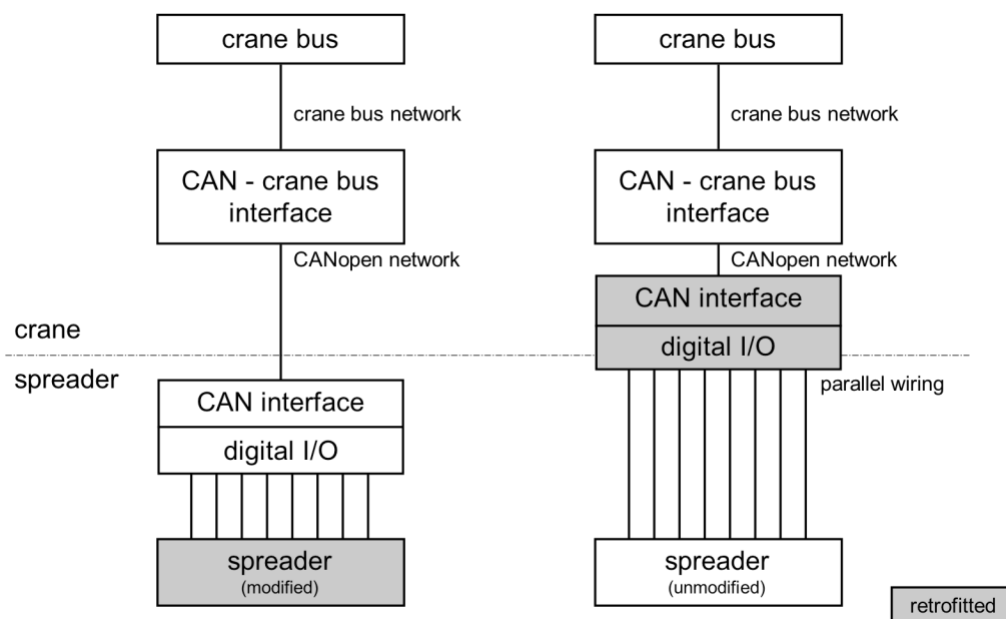


Figure 3 – CAN or direct-wired connection between a new crane and an old spreader

6 Direct-wired connection between an old crane and an old spreader

6.1 General

Figure 4 shows the direct-wired connection between an old crane and an old spreader.

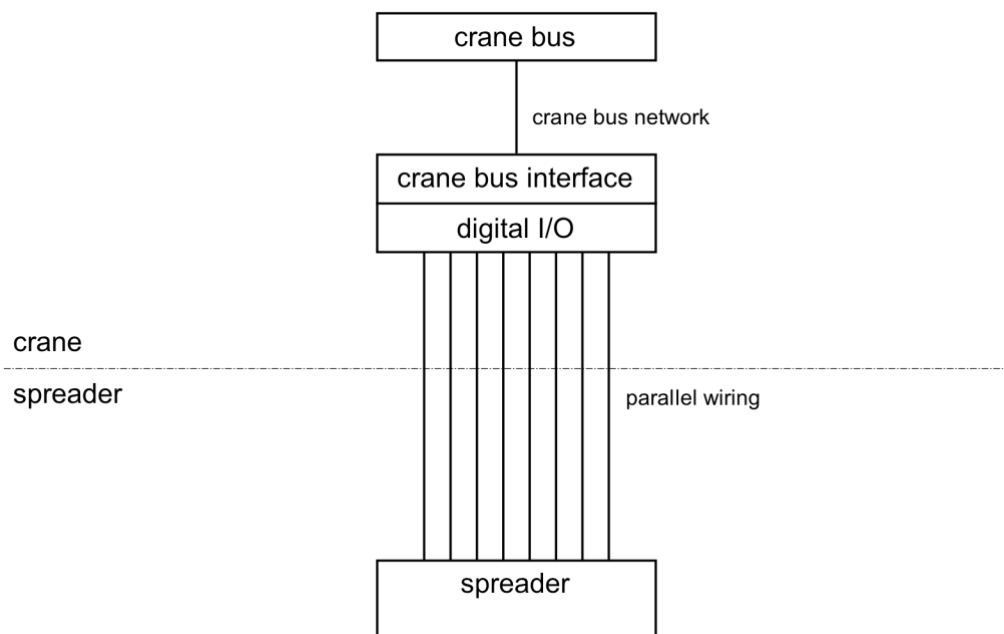


Figure 4 – Direct-wired connection between an old crane and an old spreader

6.2 Hardware compatibility

Table 1 shows the worst-case hardware configuration for connection between an old crane and an old spreader.

Table 1 – Worst-case hardware configuration for connection between an old crane and an old spreader

Hardware configuration
CAN cable length between crane and spreader up to 200 m: - Unshielded multicore cable - High-energy transmission in same cable
Up to two slip rings: Slip rings for CAN transmission not separated from energy transmission
Up to four connectors

6.3 Manufacturer-specific transmission with a boosted CANopen network

If the normal physical layer as defined in /ISO11898-2/ is not suitable for the electric environment of the crane/spreader application, a boosted physical layer can be used that is connected to the CAN networks by means of converters.

Figure 5 shows the manufacturer-specific transmission with a boosted CAN physical layer.

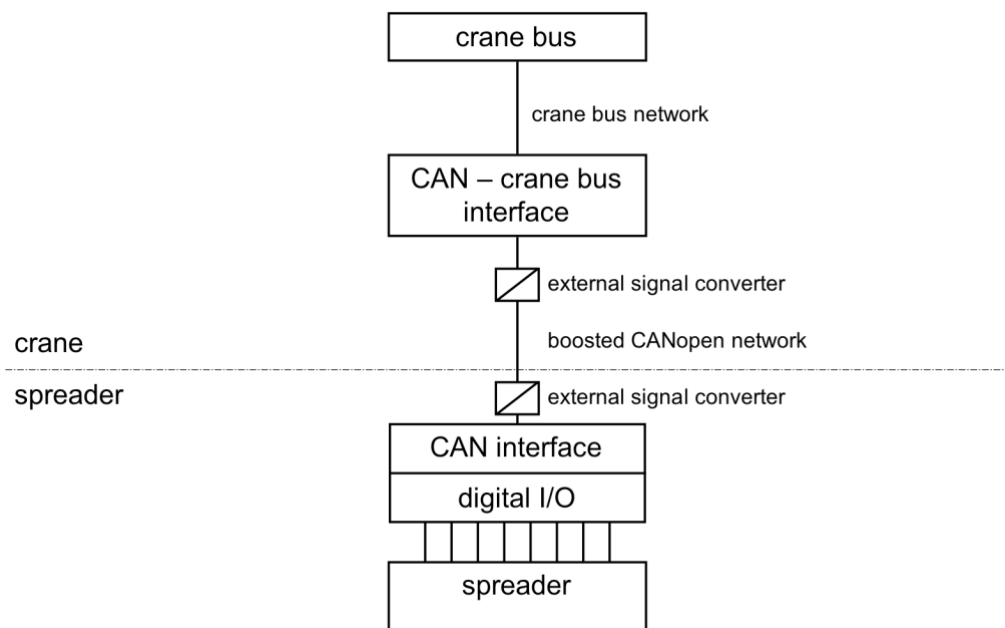


Figure 5 – Manufacturer-specific transmission with a boosted CANopen network